PERMANENCE OF THE RIGHT EXACTNESS PROPERTY (*) ALONG SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of the property (*) via the tangent functor TF. Same chapter, the definition of (*) in the statement and Lemma lemma-tangent-functor for TF. Fixes what has to be proved for each of the three parts.

Lemma 0.1. *Let A be a ring. For F a module-valued functor on Alg_A say $(*)$ holds if for all $B \in \mathrm{Ob}(Alg_A)$ the functor $TF(B, -)$ on B-modules transforms a short exact sequence of B-modules into a right exact sequence. Let $0 \to F \to G \to H \to 0$ be a short exact sequence of module-valued functors on Alg_A.*

(1) *If $(*)$ holds for F, G then $(*)$ holds for H.*
(2) *If $(*)$ holds for F, H then $(*)$ holds for G.*
(3) *If $H' \to H$ is morphism of module-valued functors on Alg_A and $(*)$ holds for F, G, H, and H', then $(*)$ holds for $G \times_H H'$.*

Proof. Let B be given. Let $0 \to N_1 \to N_2 \to N_3 \to 0$ be a short exact sequence of B-modules. Part (1) follows from a diagram chase in the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & TF(B, N_1) & \longrightarrow & TG(B, N_1) & \longrightarrow & TH(B, N_1) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & TF(B, N_2) & \longrightarrow & TG(B, N_2) & \longrightarrow & TH(B, N_2) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & TF(B, N_3) & \longrightarrow & TG(B, N_3) & \longrightarrow & TH(B, N_3) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0 & & & &
\end{array}$$

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with exact horizontal rows and exact columns involving TF and TG. To prove part (2) we do a diagram chase in the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & TF(B, N_1) & \longrightarrow & TG(B, N_1) & \longrightarrow & TH(B, N_1) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & TF(B, N_2) & \longrightarrow & TG(B, N_2) & \longrightarrow & TH(B, N_2) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & TF(B, N_3) & \longrightarrow & TG(B, N_3) & \longrightarrow & TH(B, N_3) & \longrightarrow & 0 \\
 & & \downarrow & & & & \downarrow & & \\
 & & 0 & & & & 0 & &
\end{array}$$

with exact horizontal rows and exact columns involving TF and TH. Part (3) follows from part (2) as $G \times_H H'$ sits in the exact sequence $0 \to F \to G \times_H H' \to H' \to 0$. $\square$